Mission Analysis Tool – Approach, Validation & Limitations

Mechanics & Thermodynamics of Propulsion (MECE 4305) – companion to Lecture 10

David E. Tew, Ph.D.

Columbia University ▪ david.tew@columbia.edu

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Outline

1. Why: design-roadmap steps 1–4
2. What the tool computes; architecture
3. Modeling approach (six slides)
4. Validation against textbooks, an independent ephemeris and flown missions
5. Utility: numbers for Lecture 10 and the HW 1 chain
6. Limitations: physics and scope
7. Next steps and how to run it

Companion documents

Technical report (`docs/report/`) and the tutorial *Orbital Mechanics for Propulsion Engineers* (`docs/tutorial/`), both in `Week 10 - Mission Analysis/` of the course repository.

Code

`sfmission` (`Week 10 - Mission Analysis/sfmission/`)
CLI `sfmission`, MCP server `sfmission-mcp`; 85 tests in `sfmission/tests/`.

Why: the First Four Steps of the Design Roadmap

Mission → required Δv → propellant fraction and I_{sp} → staging – then the engine

1. **Mission definition** – target, dates, orbits
2. **Required Δv** – launch, injection, flybys, capture, landing
3. **Propellant mass fraction → required I_{sp}**
4. **Staging / architecture**

5–10. thrust, thrust chamber, $\dot{m}$, V_e, p_c, M_e, A_e, feed system, iterate (Lectures 11–12)

The tool computes steps 1–4 and hands a `DeltaVBudget` with propellant masses and required I_{sp} to the engine-design steps.

Three uses

- Lecture-10 instrument (live porkchop of the Earth–Mars window opening *this month*)
- HW 1 reference solution and the first module of each team's project model
- Preliminary design tool exposed to agents via MCP

What It Computes

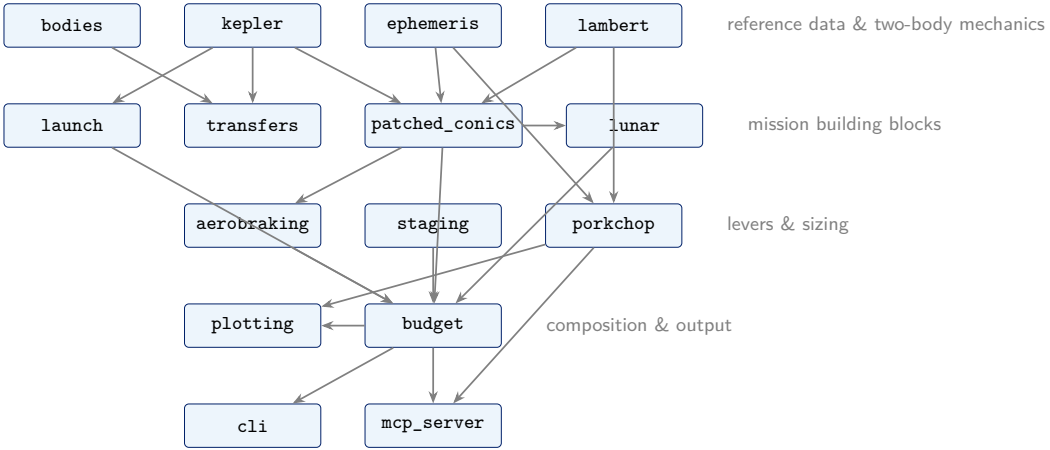
Patched conics, impulsive burns, analytic ephemerides: sizing accuracy, not navigation accuracy

Launch	Δv to orbit: $\lvert \boldsymbol{v}_{orbit} - \boldsymbol{v}_{site} \rvert$ with azimuth for inclination, loss band
Transfers	Hohmann, bi-elliptic, plane change (optimal split), phasing, round trips
Ephemeris	JPL mean elements (1800–2050; 3000 BC–3000 AD), truncated-Meeus Moon
Lambert	universal-variable, bracketed root; type I/II, prograde/retrograde
Patched conics	departure & capture burns, unpowered & powered flybys, multi-leg sequences
Porkchops	C_3, v_∞^{arr}, Δv grids; optima; window spans; multi-year scans
Earth–Moon	Hohmann estimate; Bate–Mueller–White patched conic with perilune design solver
Aero levers	aerocapture credit; per-pass aerobraking Δv; campaigns; integrated pass check
Staging	rocket equation, required I_{sp}, closed-form optimal N-stage split
Budgets	legs with bands and margins $\rightarrow$ propellant mass, ζ, required I_{sp}
Front ends	Python API, CLI (`sfmission`), 23 MCP tools (`sfmission-mcp`)

Fidelity class: patched conics, impulsive burns, analytic ephemerides – sizing accuracy, not navigation accuracy.

Architecture

Pure functions returning dataclasses; the three front ends only format



km, s, rad internally; degrees for *_deg arguments; epochs as ISO strings or Julian Date. 85 tests run in 1 s.

Approach I: Reference Data and Their Accuracy

JPL DE440 constants; JPL mean-element planets; truncated-Meeus Moon

- *GM*: JPL DE440 (planet-*system* values for outer planets); IAU radii; Laplace SOI $a(m/M)^{2/5}$
- Planets: JPL *Keplerian Elements for Approximate Positions* – six elements + secular rates, J2000 ecliptic
- Moon: truncated Meeus Ch. 47 (32 lon/dist, 20 lat terms), precessed to J2000
- "Earth" = Earth–Moon barycenter unless corrected

vs. astropy	angle [″]	rel. dist.	v [m/s]
Inner planets	< 35	$< 1.5 \times 10^{-4}$	< 5
Jupiter	< 340	$< 4 \times 10^{-4}$	< 28
Saturn	< 560	$< 1.5 \times 10^{-3}$	< 41
Uranus/Neptune	< 110	$< 1 \times 10^{-3}$	< 16
Moon	< 40	$< 40\,\mathrm{km}$	–

Consequence for design: v_∞ errors $\sim 1\,\mathrm{m/s}$ at Earth, $\sim 10\,\mathrm{m/s}$ at Saturn – far below budget margins.

Refs: Standish, JPL approximate planetary elements; Park et al. 2021 (DE440); Meeus, Astronomical Algorithms Ch. 47

Approach II: Lambert Arcs and Porkchop Maps

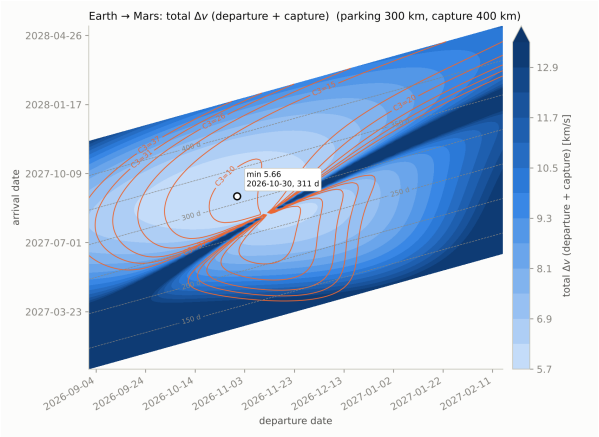
Dated transfers between ephemeris positions; the launch window as a map

Universal-variable Lambert (BMW/Curtis):

$$F(z) = \left[\frac{y}{C}\right]^{3/2} S + A\sqrt{y} - \sqrt{\mu}\,\Delta t = 0$$

- F monotonic on the single-rev branch $\Rightarrow$ bracket + Brent: never diverges
- type I ($< 180^\circ$) and type II fall out of the prograde geometry
- 55 µs per solve; 110^2 grid in $\sim$1 s
- each cell: C_3, v_∞^{arr}, Δv_{dep}, Δv_{cap}, DLA

Earth→Mars 2026/27 optimum: depart 2026-10-30, 311 d, $C_3 = 9.2$, $\Delta v = 5.66$ km/s (300 km parking → 400 km Mars orbit).



Approach III: Patched Conics

Departure and capture burns, unpowered and powered flybys

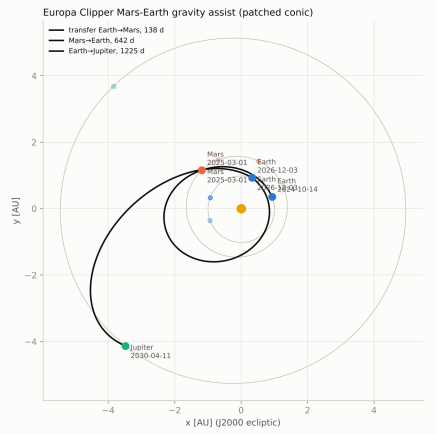
Departure / capture (tangential periapsis burns)

$$\Delta v_{dep} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - \sqrt{\frac{\mu}{r_p}}, \quad \Delta v_{cap} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - v_p^{orbit}$$

Unpowered flyby

$$e = 1 + \frac{r_p v_\infty^2}{\mu}, \quad \delta = 2 \arcsin \frac{1}{e}, \quad |\Delta v_{helio}| = 2 v_\infty \sin \frac{\delta}{2}$$

Powered flyby (MGA model): shared r_p with $\frac{1}{2}\delta_{in} + \frac{1}{2}\delta_{out} = \angle(\boldsymbol{v}_\infty^{in}, \boldsymbol{v}_\infty^{out})$, speed mismatch burned at periapsis; infeasible if $r_p < r_{min}$.



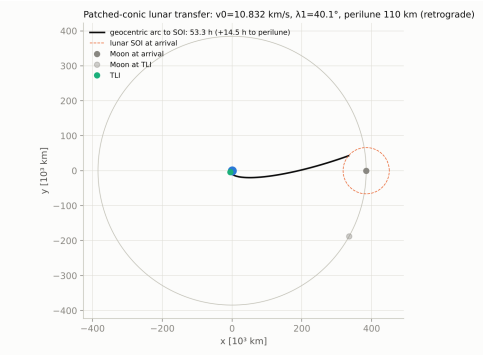
Europa Clipper dates: C_3 41 vs. 77 direct; injection 4.95 vs. 6.30 km/s; powered corrections 0.13 + 0.13 km/s at Mars and Earth.

Approach IV: Earth–Moon Patched Conic

Bate–Mueller–White §7.4, implemented with vectors; SOI transit time included

- Geocentric ellipse from (r_0, v_0, ϕ_0) to the SOI point set by λ_1
- $\boldsymbol{v}_2 = \boldsymbol{v}_1 - \boldsymbol{v}_{Moon} \rightarrow$ selenocentric hyperbola (e, r_p, v_p)
- **Vector implementation**: sense of motion from $\mathrm{sign}(h_2)$, no trig sign ambiguities (scalar relations reproduced in tests)
- Adds the **SOI transit time** (14–18 h) – essential to compare with flown data
- Design solver: λ_1 for given v_0 (or v_0 for λ_1) hitting a perilune altitude; prograde and retrograde roots

Hohmann estimate: TLI 3.135, LOI 0.82 km/s; 3-day transfer: TLI 3.157, LOI 0.889 km/s.



Approach V: Launch Δv and the Aero Levers

Site-rotation credit and loss band; aerocapture credit; per-pass aerobraking Δv

Launch

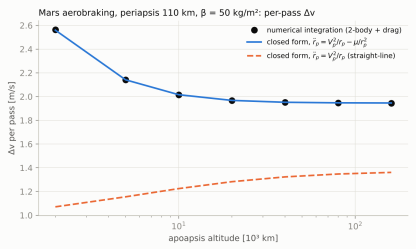
$$\Delta v = |\boldsymbol{v}_{orbit} - \boldsymbol{v}_{site}| + \Delta v_{grav} + \Delta v_{drag} + \Delta v_{steer}$$

with $\boldsymbol{v}_{site} = \omega R \cos\varphi\, \hat{\boldsymbol{e}}_E$ and $\cos i = \cos\varphi \sin Az$. KSC, 300 km, due east:
$7.726 - 0.409 + (1.15 \ldots 2.05) = 8.47 - -9.37$ km/s.

Aerobraking, per pass (exponential atmosphere, parabolic $r(t)$ about periapsis)

$$\Delta v_{pass} = \frac{\rho_p V_p^2}{2\beta}\sqrt{\frac{2\pi H}{\ddot{r}_p}}, \qquad \ddot{r}_p = \frac{V_p^2}{r_p} - \frac{\mu}{r_p^2}$$

Dropping μ/r_p^2 (straight-line fly-through) gives the Lecture-11 slide form, low by $\sqrt{V_p^2/(V_p^2 - \mu/r_p)} \to \sqrt{2}$; integration confirms the gravity-aware form to 0.2%.



Aerocapture credit at Mars ($v_\infty = 2.65$): 2.08 km/s replaced by a 0.06 km/s periapsis raise.

Approach VI: Rocket Equation, Staging, Budgets

The hand-off from mission analysis to engine design

Notes nomenclature: $\zeta = m_p/m_0$, $\sigma_i = m_{s,i}/(m_{s,i} + m_{p,i})$, $\pi_i = m_{0,i+1}/m_{0,i}$, $c_i = g_0 I_{sp,i}$, $R_i = e^{\Delta v_i / c_i}$ (ε reserved for area ratio)

$$\frac{m_0}{m_L} = \prod_i \frac{1}{\pi_i} = \prod_i \frac{R_i(1-\sigma_i)}{1 - R_i\sigma_i}, \qquad I_{sp}{}^{req} = \frac{\Delta v}{g_0 \ln[1/(1-\zeta)]}$$

Optimal split for fixed $\sum \Delta v_i$ (one scalar root in the multiplier λ):

$$R_i = \frac{\lambda c_i - 1}{\lambda c_i \sigma_i}, \qquad \sum_i c_i \ln R_i = \Delta v_{tot}$$

DeltaVBudget

ordered legs with nominal/low/high, kind (propulsive / aero / margin), optional stage I_{sp}

- composers: dated interplanetary (launch, injection, flybys, capture or aerocapture, landing, margin); lunar (TLI, LOI, descent, ascent, TEI)
- backward propellant sizing; ζ and required I_{sp} – the hand-off to engine design

Validation: Textbook Cases, an Independent Ephemeris, Flown Missions

All in `sfmission/tests/` (85 tests, $\sim$1 s)

Lambert, Curtis Ex. 5.2; 20 closure geometries	$\boldsymbol{v}_1, \boldsymbol{v}_2$ to 1×10^{-4} km/s; closes to 1×10^{-8} km
Earth$\rightarrow$Mars Hohmann (Curtis Ex. 8.4–8.6)	v_∞ 2.943/2.648, Δv 3.590 km/s, phase 44.3°, synodic 779.9 d
LEO$\rightarrow$GEO; +28.5° plane change	3.893 km/s; 4.23 km/s
Mars 2020 actual dates	path 470.9 Mkm vs. $\approx$471 Mkm (JPL); C_3 14.5; v_∞^{arr} 2.55 km/s
Voyager 2 E–J–S dates	Jupiter turn 97°; r_p 620,000 vs. $\approx$570,000 km; powered fix 0.05 km/s; C_3 105
Apollo 11 published TLI state	TLI$\rightarrow$perilune 68–73 h vs. 73.0 h; LOI 0.89–0.92 vs. 0.891 km/s
Aerobraking pass vs. integration	1.952 vs. 1.951 m/s
Ephemeris vs. astropy; Mars oppositions	table on slide 5; longitudes agree $< 1.5°$
Staging identities, optimal split	exact

Lecture point from Apollo 11: the two published TLI rows differ by 0.04 $\mathrm{km}^2/\mathrm{s}^2$ in energy, and lunar TOF moves $\sim$5 h per 0.1 $\mathrm{km}^2/\mathrm{s}^2$ – the result is as good as the altitude datum.

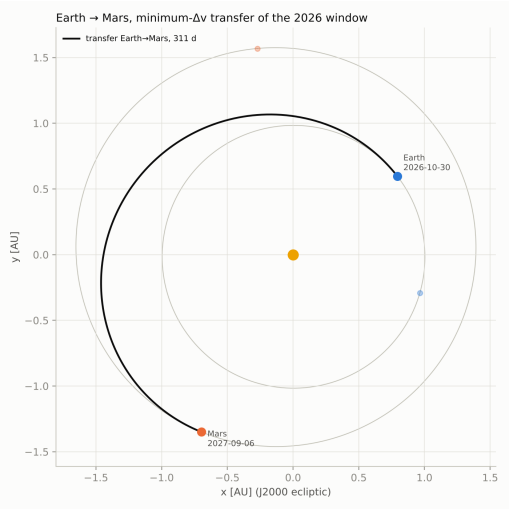
Refs: Curtis, Orbital Mechanics for Engineering Students; NASA/JPL Mars 2020 press kit; Apollo 11 Mission Report trajectory tables

Utility: Numbers for Lecture 10

The Δv levers, quantified for the Earth–Mars window opening now

Lever (Mars $v_\infty = 2.65$ km/s)	without	with
Aerocapture, 400 km orbit	2.08 km/s	0.06 km/s
Aerobraking 110×40,000→400 km	2.08 km/s	0.92 km/s, 152 d
Mars–Earth assist to Jupiter	C_3 77, 2.7 yr	C_3 41, 5.5 yr
KSC credit, east vs. polar	0.41 km/s	−0.01 km/s

- Earth–Mars phase angle on 11/12: 56° vs. 44.3° Hohmann requirement
- Type II (311 d, 5.66 km/s) vs. type I (260 d, 6.02 km/s) in this window
- Next windows: 2028-11, 2030-12/2031-02, 2033-03/04 (type II / type I pairs)



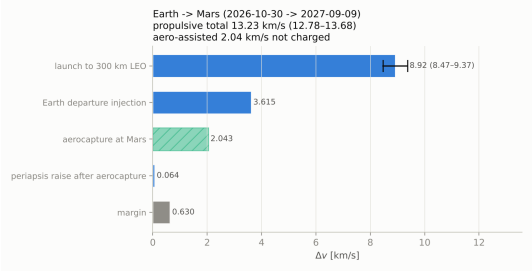
Utility: the HW 1 Reference Chain

Design-roadmap steps 1–4 end to end for a Mars orbiter in the 2026 window

Mars orbiter, 2026 window, KSC:

1. launch band 8.47–9.37 km/s
2. dated legs at the porkchop optimum
3. budgets: post-launch Δv **6.39** (propulsive) vs. **4.31** km/s (aerocapture)
4. required I_{sp} at $\zeta = 0.7$: 541 s vs. 365 s
5. 1000 kg final mass, 320-s engine $\rightarrow$ 7,653 kg injected
6. two-stage launcher (I_{sp} 300/450 s, σ 0.07/0.10) for 12.5 km/s: gross 695 t, optimal split 4.8/7.7 km/s

Teams swap mission, dates and engines; the `DeltaVBudget` feeds the Week 11–12 thermochemistry and engine modules.



Limitations I: Physics

Deliberate modeling choices and what they cost

- **Patched conics**: no third-body gravity on a conic, sharp SOI. Few-% error in planetocentric burns for planetary missions; hours of TOF and tens of m/s of LOI for the Moon (Apollo 11: 0–7%).
- **Impulsive burns**: finite-burn losses of low-T/W stages not charged; launch losses are a user band, not a trajectory (the ascent tool computes them).
- **Ephemeris**: arcsec–arcmin mean elements; Earth = EM barycenter by default. Fine for Δv, not for targeting.
- **Circular assumptions**: Moon orbit circular at 384,400 km (perigee–apogee $\pm$21,000 km $\Rightarrow$ $\pm$10 m/s TLI, hours of TOF); quick estimates use circular coplanar planets.
- **Plane/DLA**: 3-D legs, but no plane-change penalty for high-DLA departures or parking-orbit incompatibility; DLA reported only.
- **Lambert**: single revolution; 180° transfers undefined (NaN ridge).
- **Powered flyby**: shared-periapsis tangential-burn model is a lower bound; flyby dates are inputs, not optimized.

Limitations II: Scope

What is charged as a number only, and what is not modelled at all

Aero-assist

- aerocapture charged as a Δv credit only: no corridor, heating, g-load, TPS mass, guidance
- single-scale-height atmosphere fitted to 90–150 km; constant β; Mars density varies $\times 2 \Rightarrow$ pass count and duration do too

Staging

- σ is an input; no T/W, tank or residual models; σ_k independent of stage size

Reference values, not computed

- lunar descent/ascent 2.07/1.86 km/s (Apollo LM); Mars landing Δv user-supplied

Not covered

- low-thrust / electric trajectories
- launch-vehicle performance curves (C_3 vs. payload)
- navigation and statistical Δv, environments, operations

Next Steps and How to Run It

Planned extensions, in priority order for the course

Planned (priority order)

1. low-thrust legs: Edelbaum Δv, power-limited sizing (HW 3b, EP teams)
2. DE440 option + perturbed propagator to quantify patched-conic error on request
3. multi-revolution Lambert
4. launch-vehicle performance curves
5. coupling to the ascent-trajectory optimizer (computed launch band)

```
pip install -e ".[dev,mcp]"      # repo root
pytest sfmission/tests    # 85 tests

from sfmission import *
pc = porkchop("Earth","Mars","2026-09-01","2027-02-15
     ",120,450)
print(pc.summary())
b, m = interplanetary_budget("Earth","Mars",
        "2026-10-30","2027-09-06", aerocapture=True)
print(b.table()); b.size_propellant(1000, 320)

sfmission lunar --tof 72 --plot lunar.png
sfmission-mcp     # {"mcpServers":{"sfmission":
                  #   {"command":"sfmission-mcp"}}}
```

Code, tests, examples, figures, report, tutorial and these slides: `Week 10 - Mission Analysis/` in the course repository.